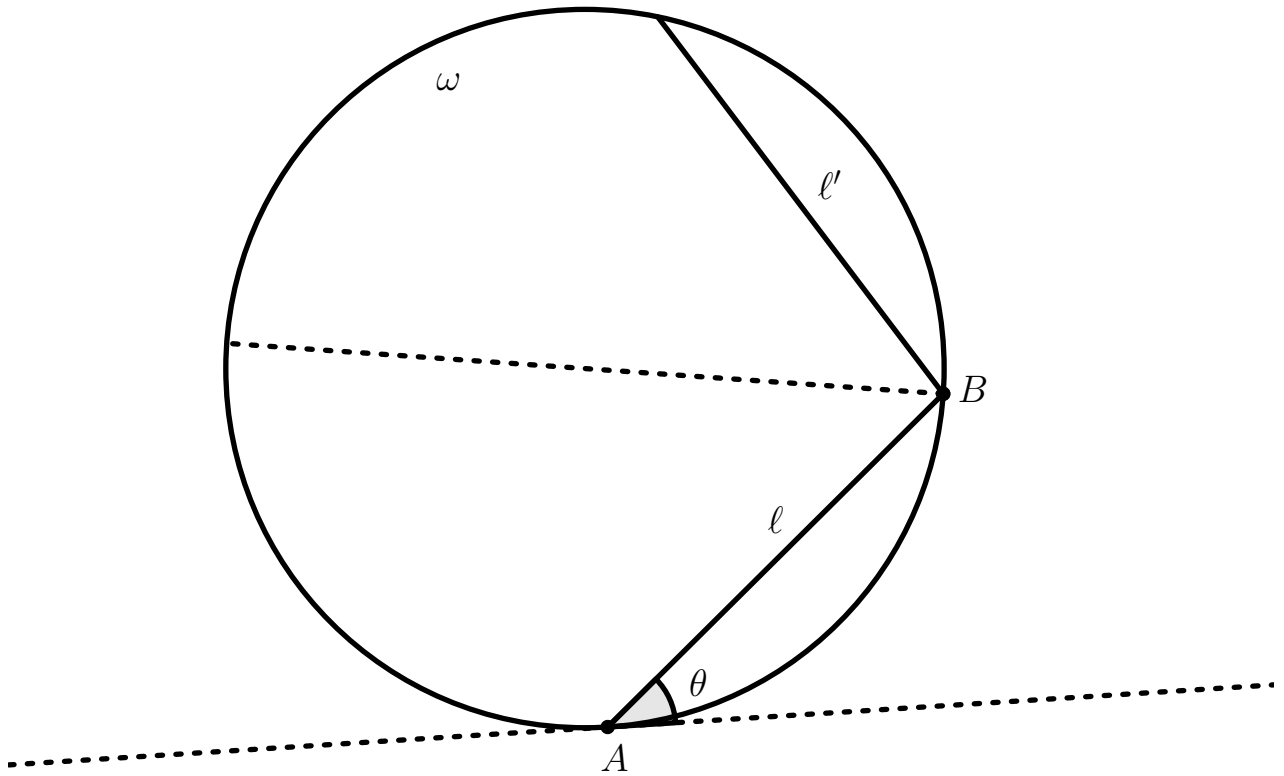


Reflective Circles

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A laser starts at point A on the circumference of a circle, ω . The laser moves along a chord from point A to some other point B on ω , labeled ℓ . The angle between the line tangent to the circle at point A and ℓ is θ . When the laser reaches point B , it bounces off the circle. It then moves along ℓ' , which is the reflection of ℓ over the diameter of ω that passes through point B .



Problem 1: Assuming this process continues indefinitely, for which values of θ does the laser's path reach point A ?

Problem 2: What can we say about the set of points on ω that the laser will never reach?